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Programming

Solving Problems with Computers

Dr. Le Van Quoc Anh
anh@ut.edu.vn

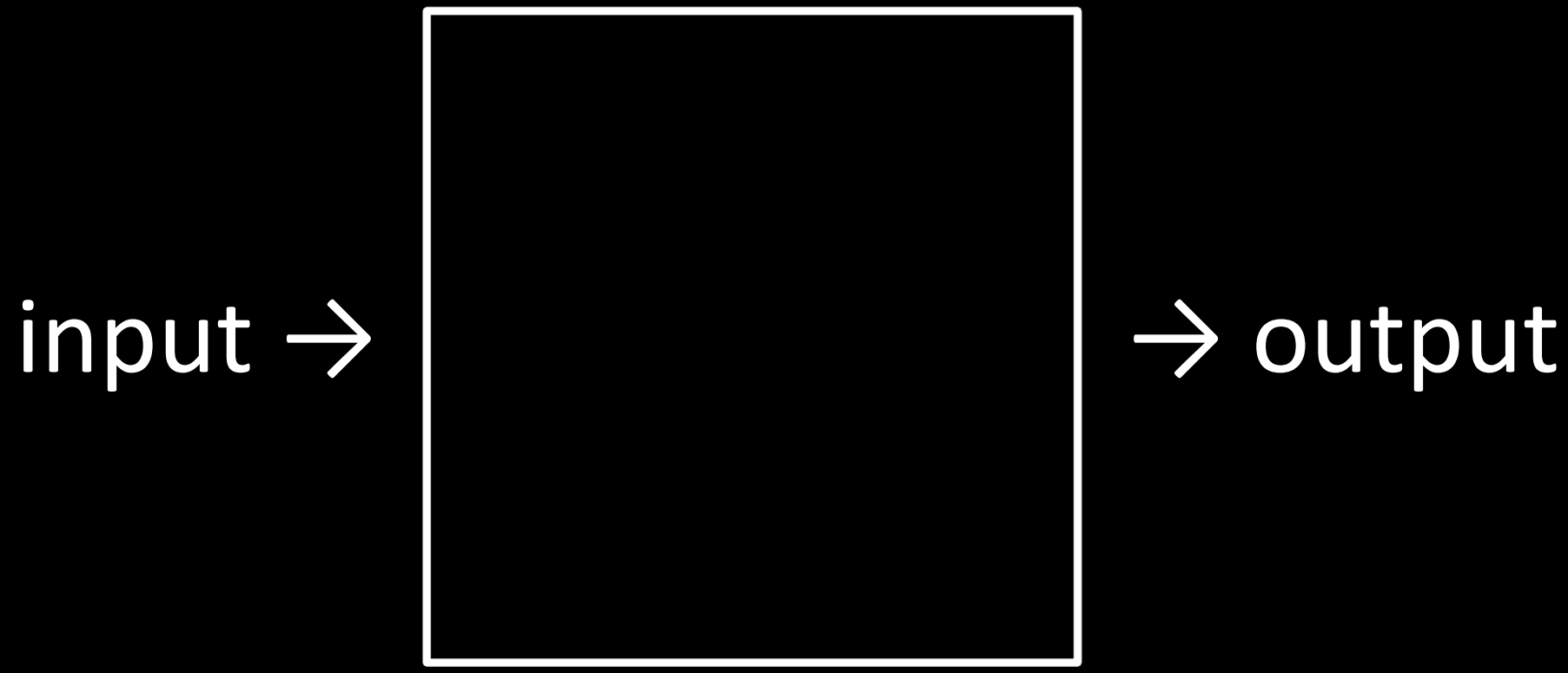
Credit: Several slides are adapted from Harvard's CS50 courses

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Programming

How to tell computers solve our problem?

Learn how to think algorithmically


input →
→ output

algorithm

code



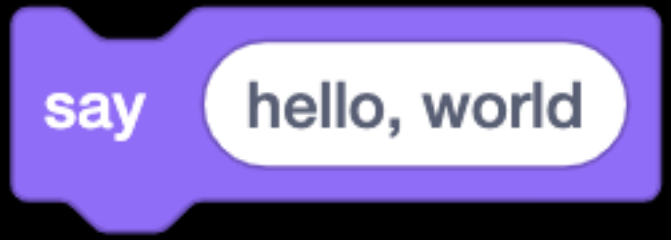

```
# This is Python code

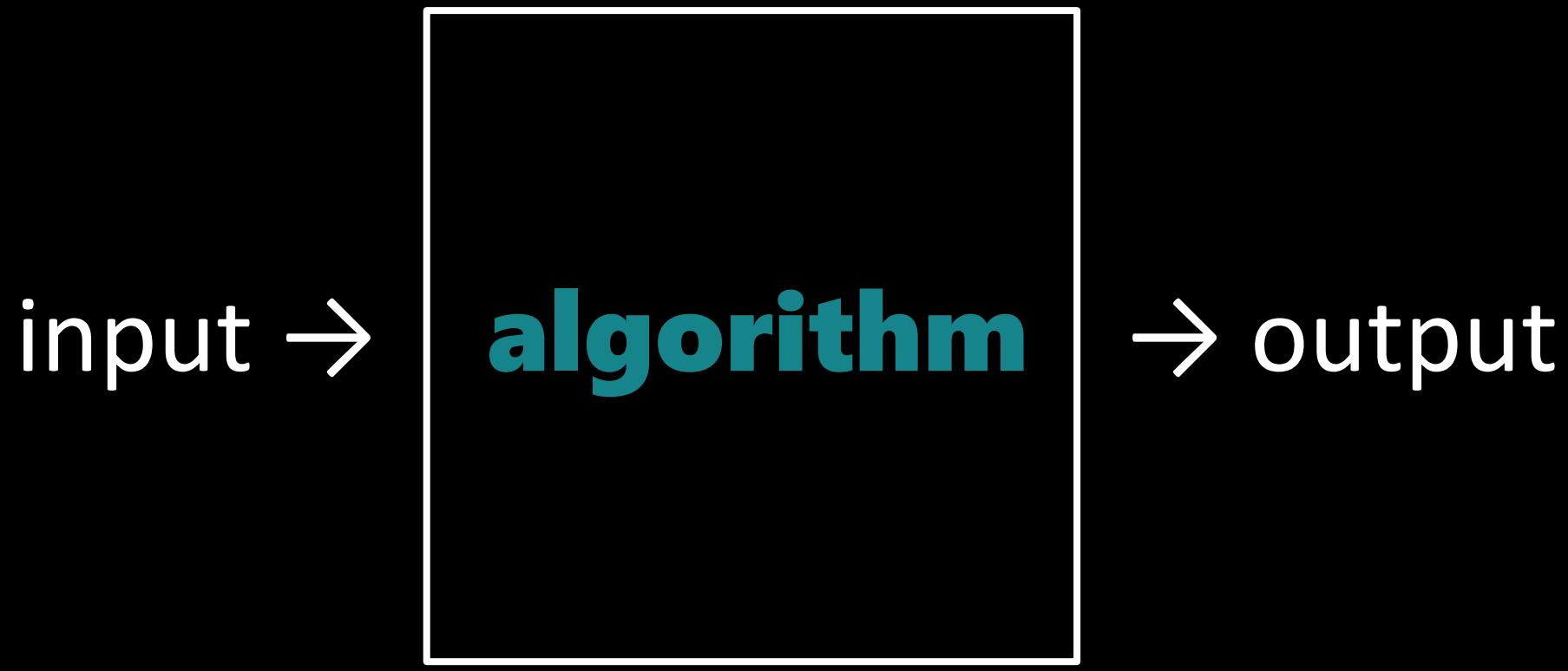
print('hello, world')
```

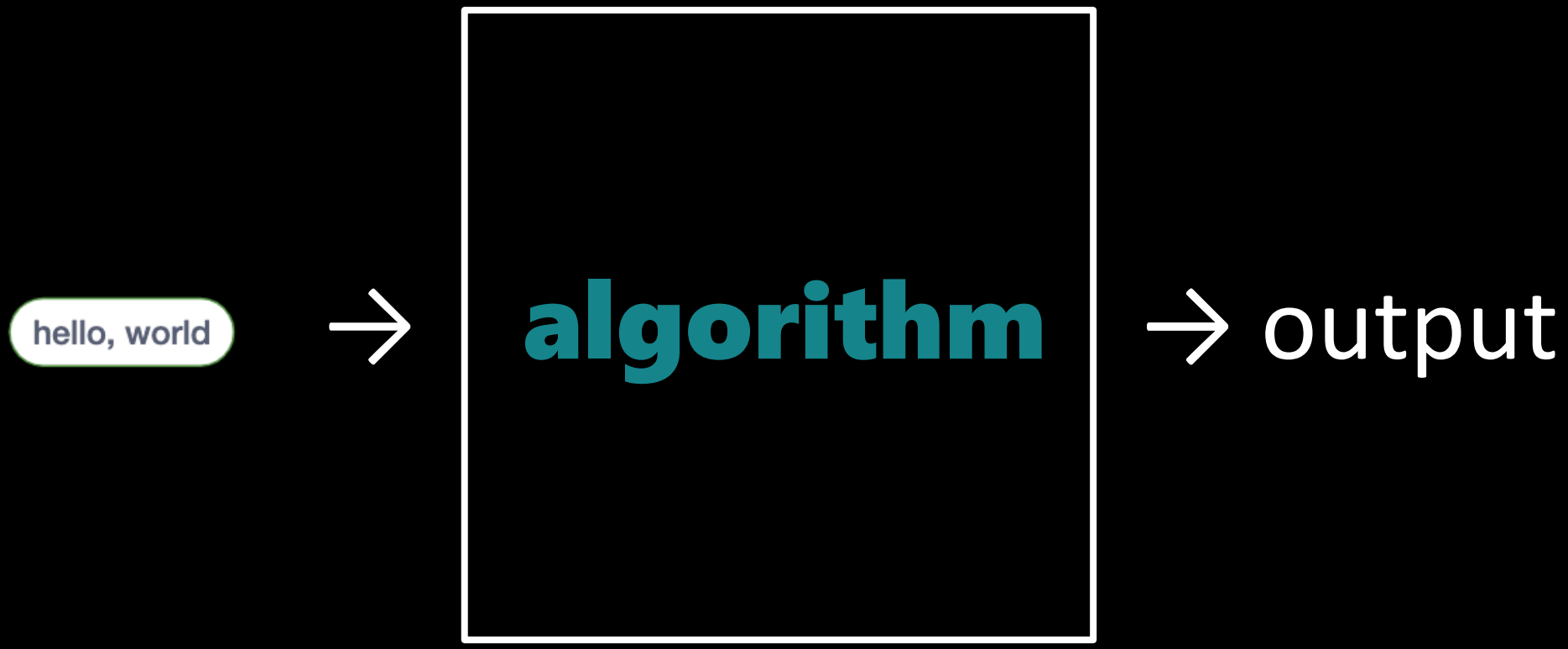
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Let's code...

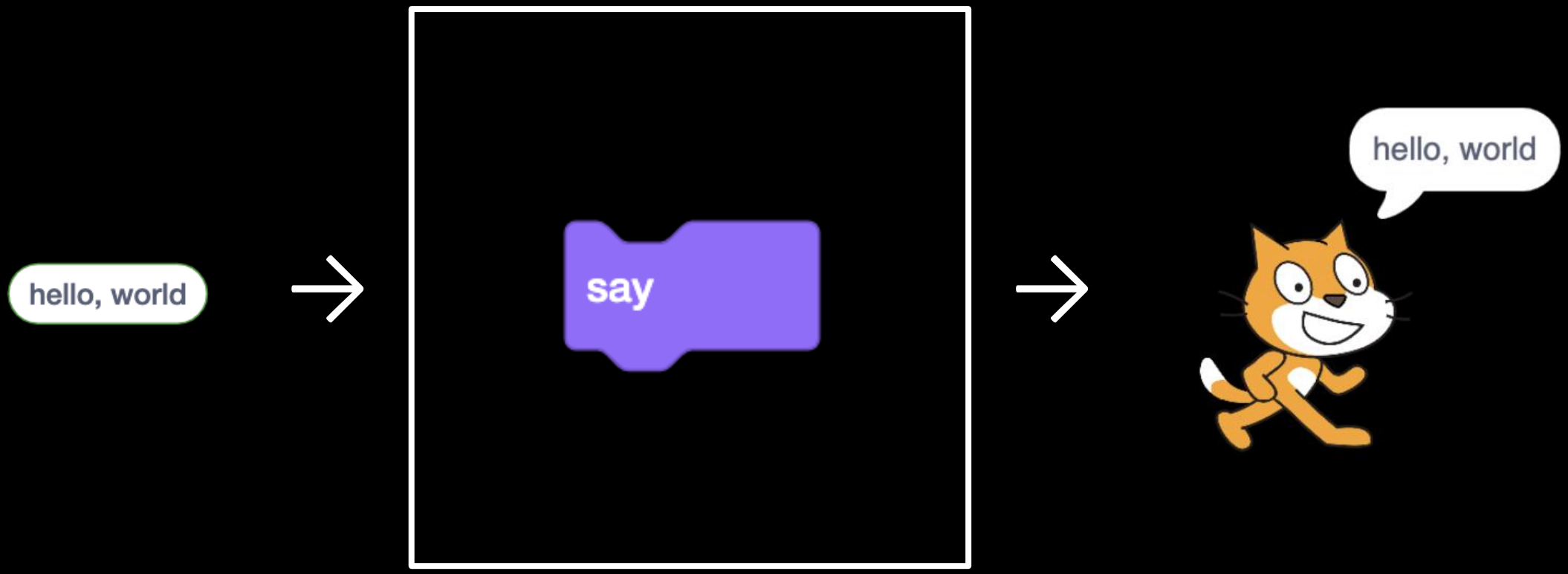
https://scratch.mit.edu
https://www.online-python.com/


say
hello, world


input →
algorithm
→ output

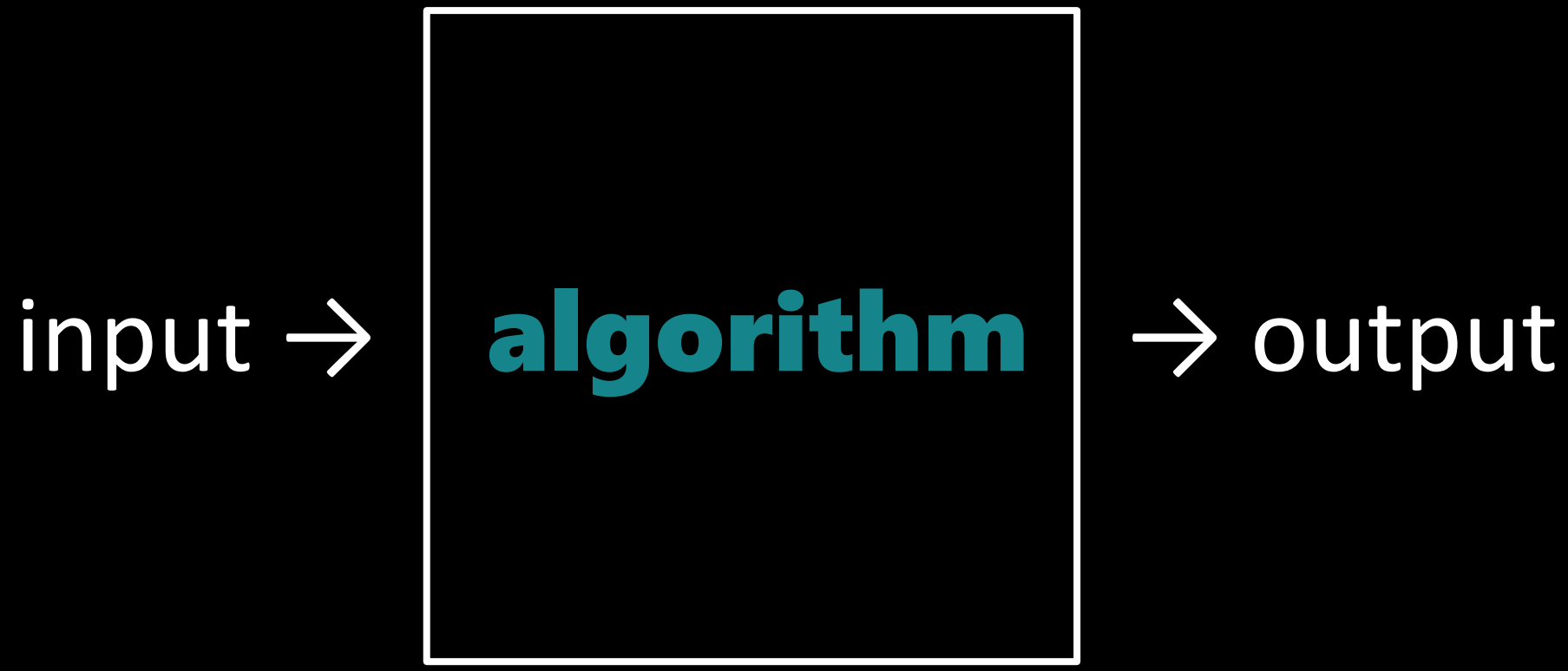

hello, world
algorithm
output

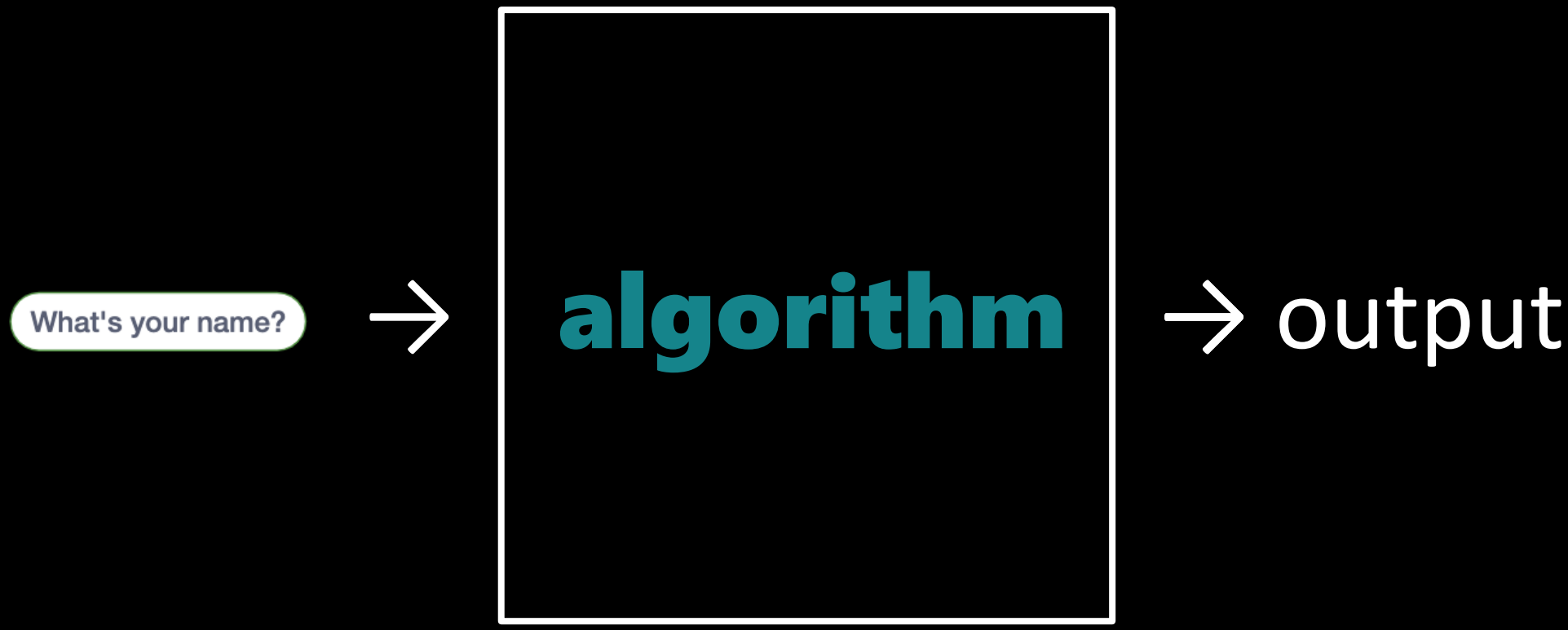

hello, world
say
output

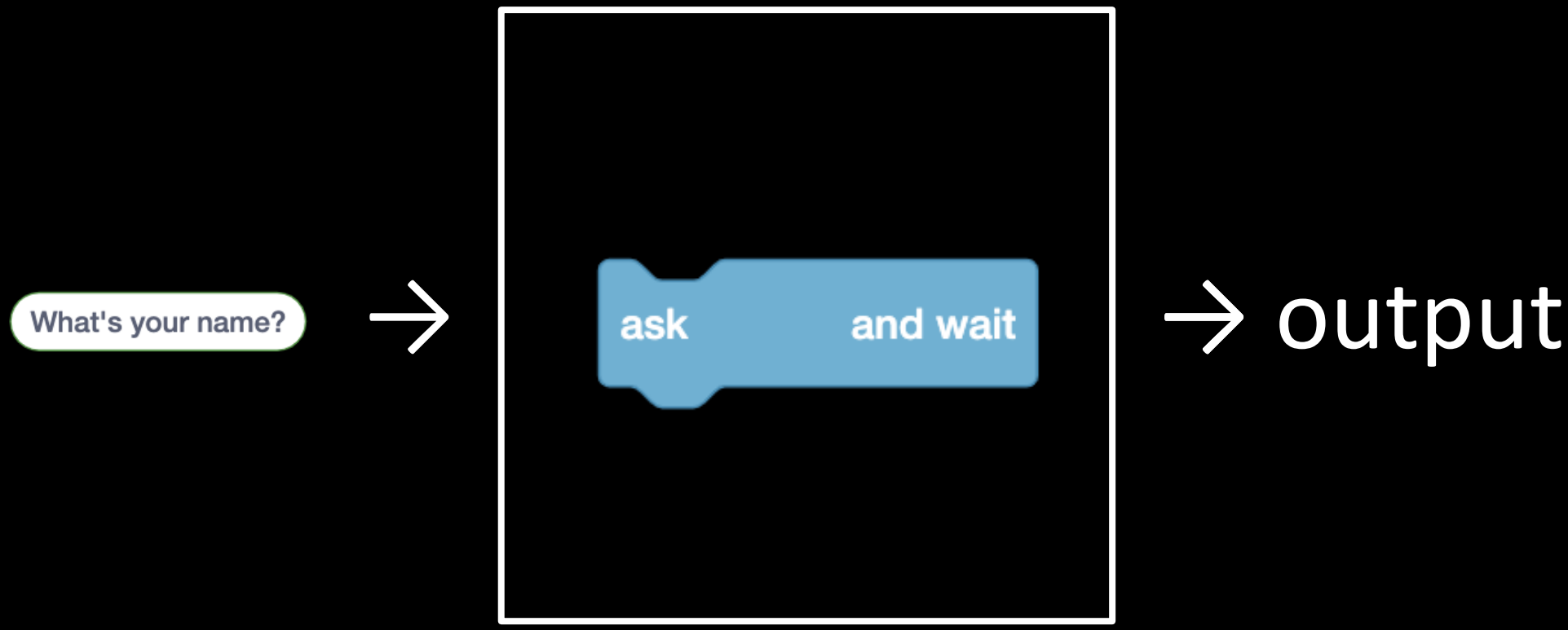

hello, world
say
hello, world

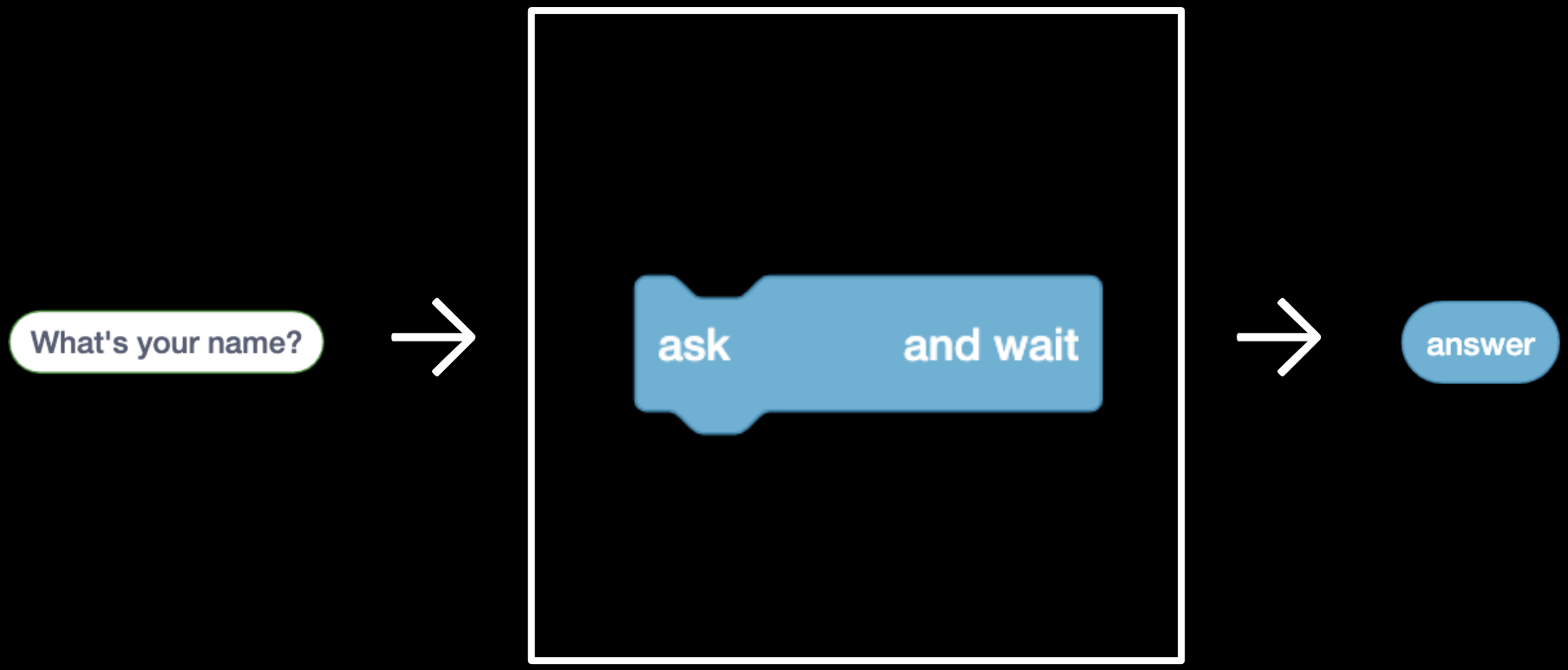
Do something more complicated


ask
What's your name?
and wait


input →
algorithm
→ output


What's your name?
algorithm
output


What's your name?
ask
and wait
output


What's your name?
ask
and wait
answer

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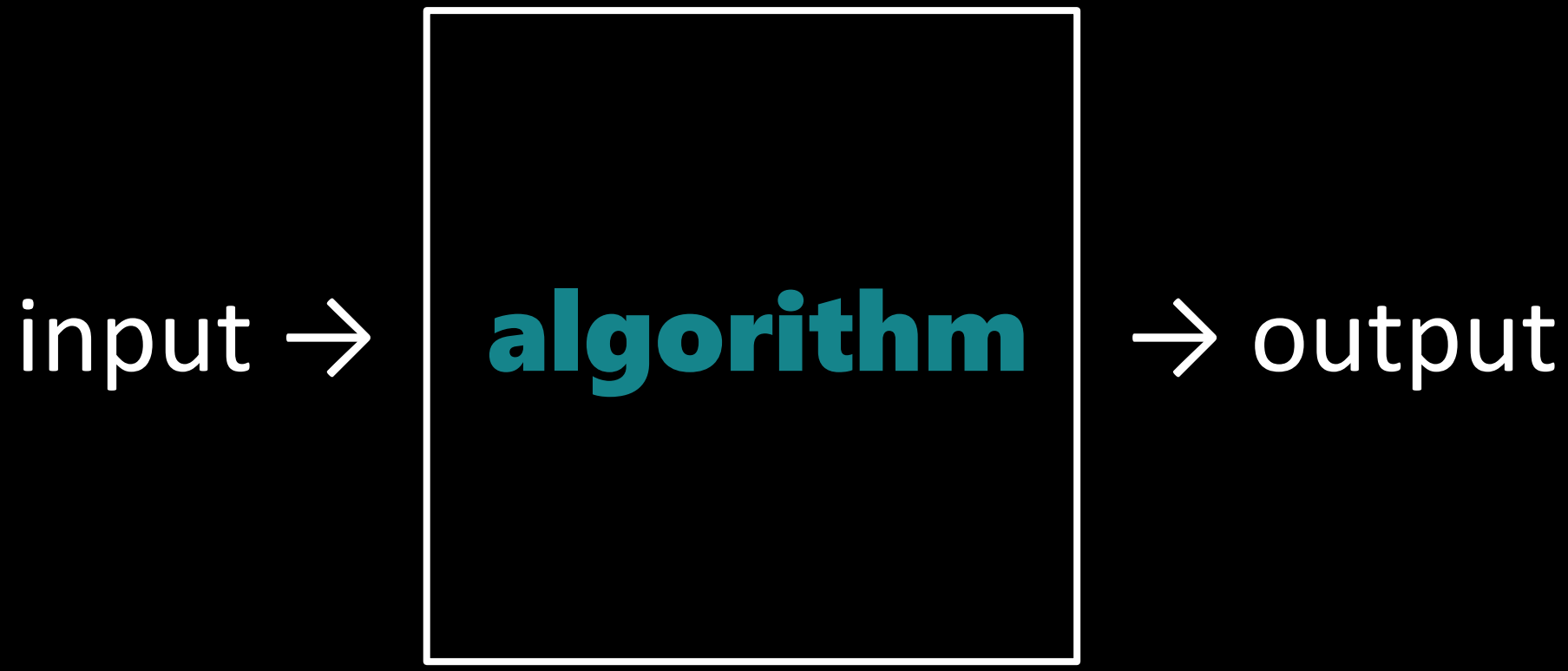
Exercise 1:

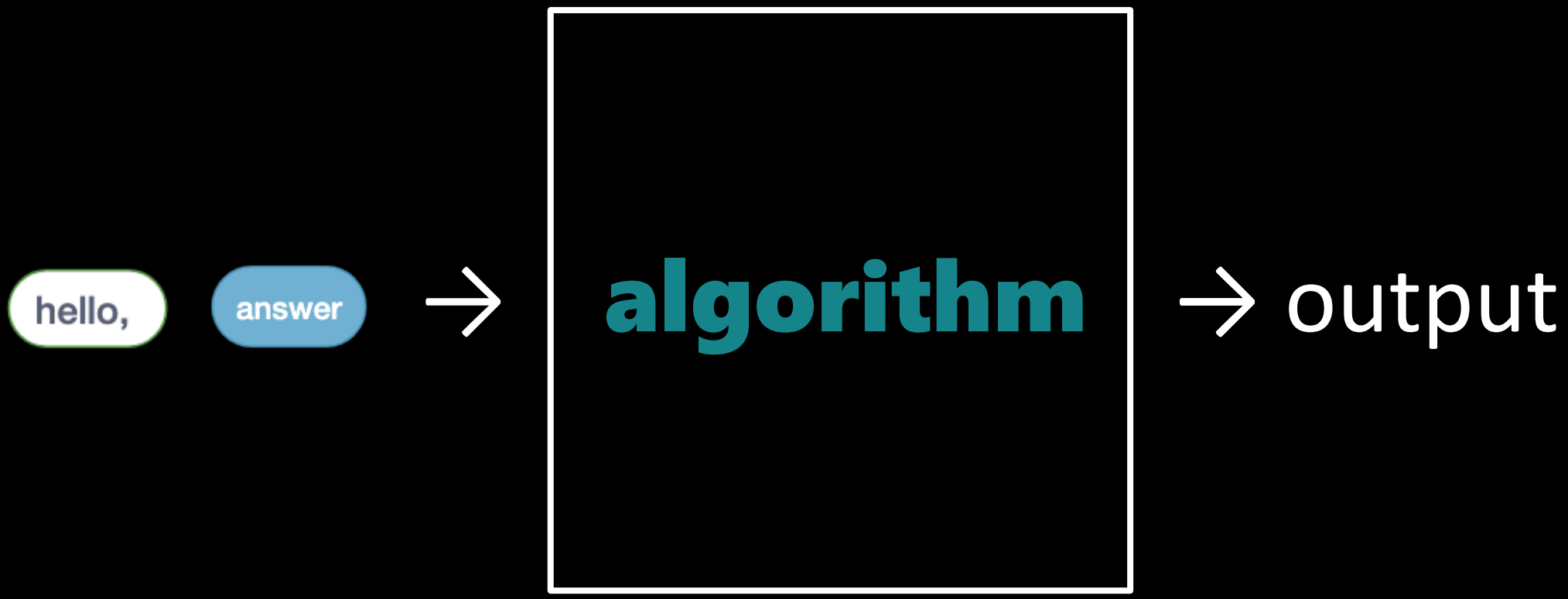
Modify the previous program as follows:

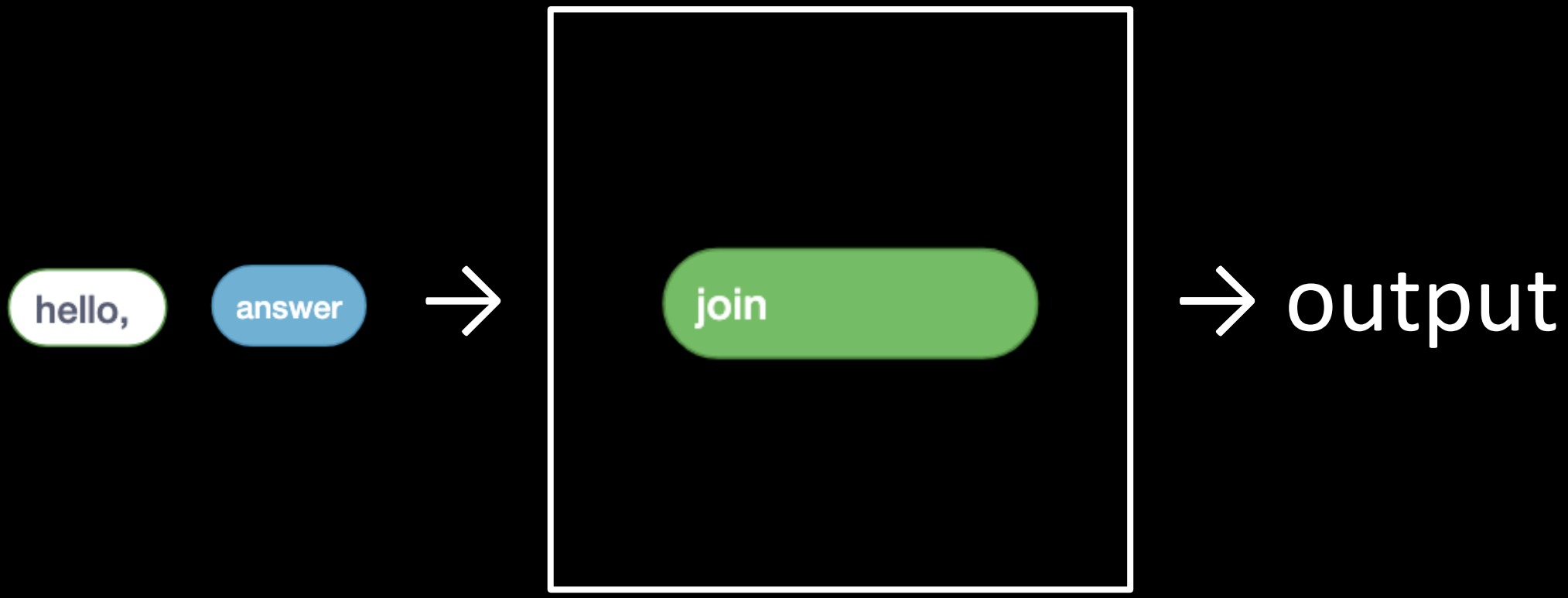
Ask the name of the user
and
let the cat say hello with the name

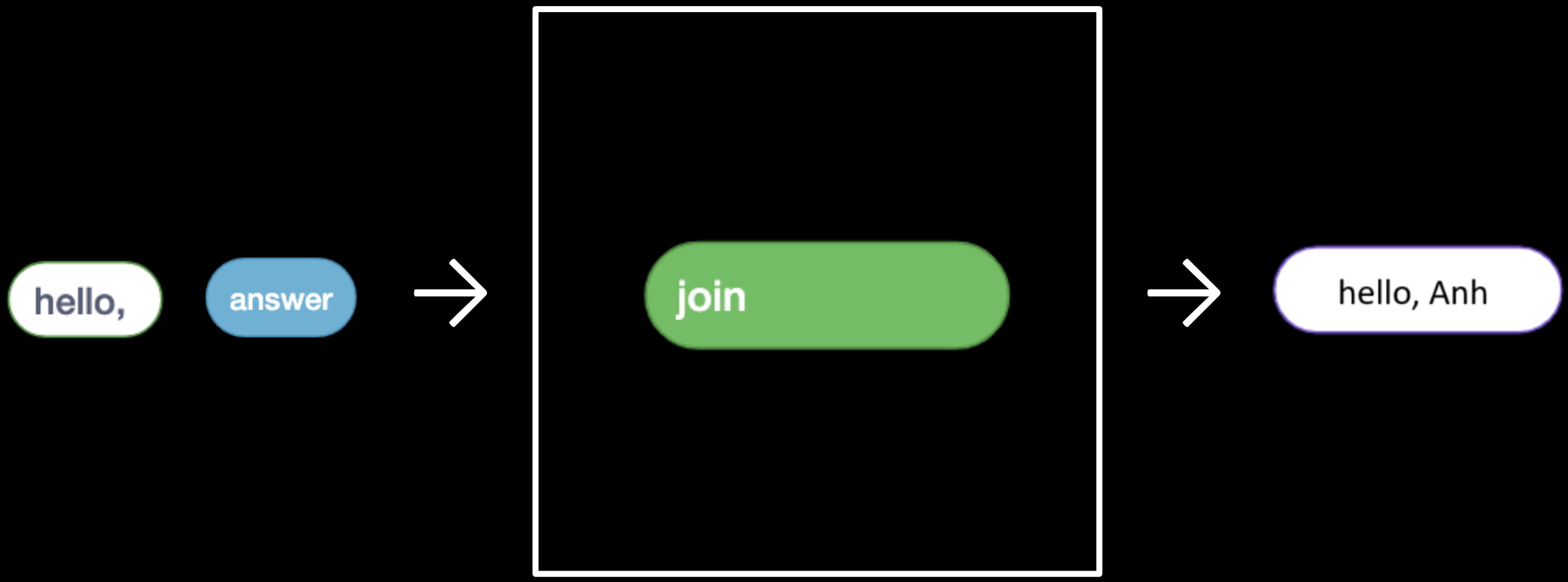



say
join
hello,
answer


input →
algorithm
→ output

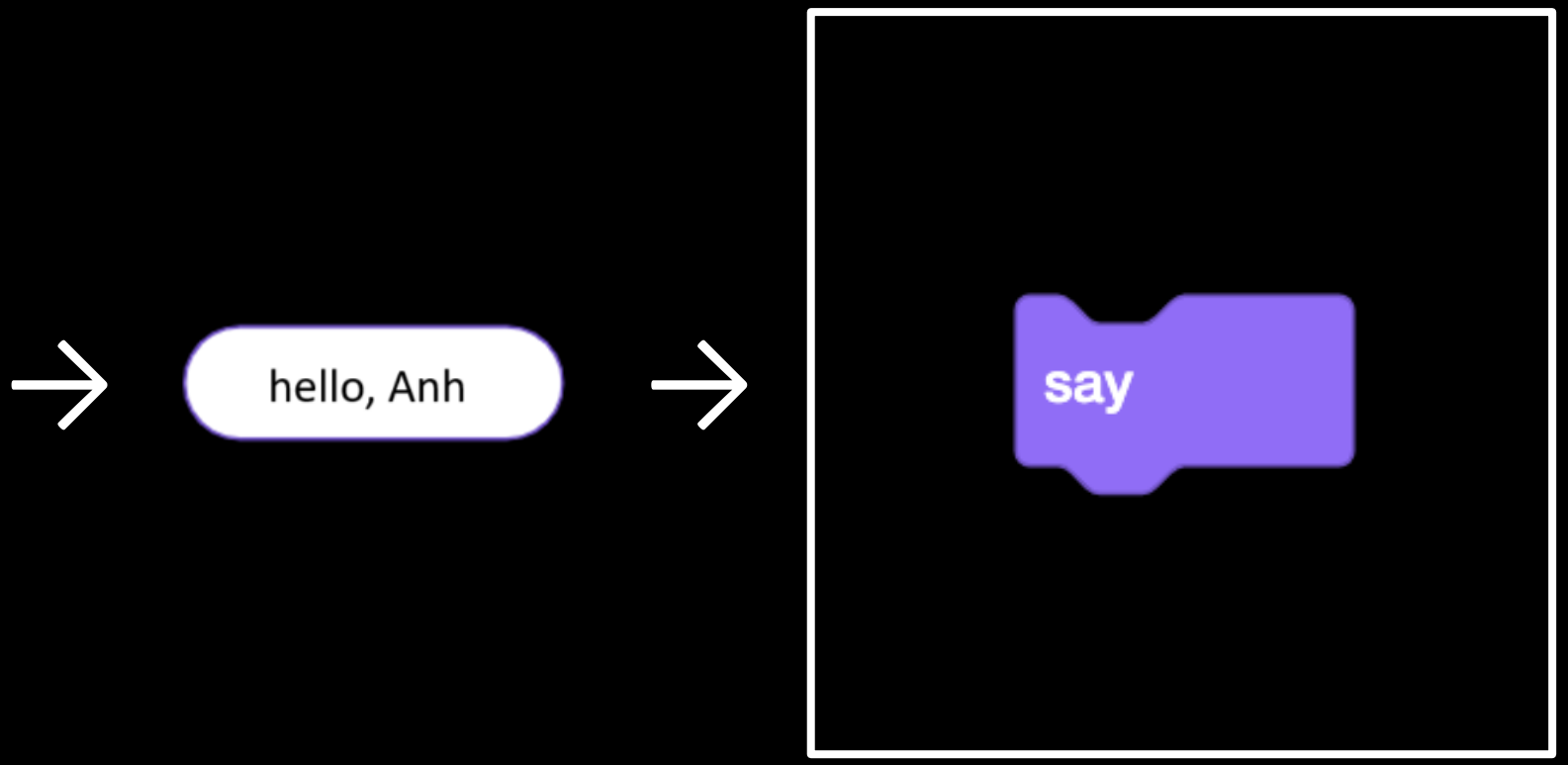

hello,
answer
algorithm
output

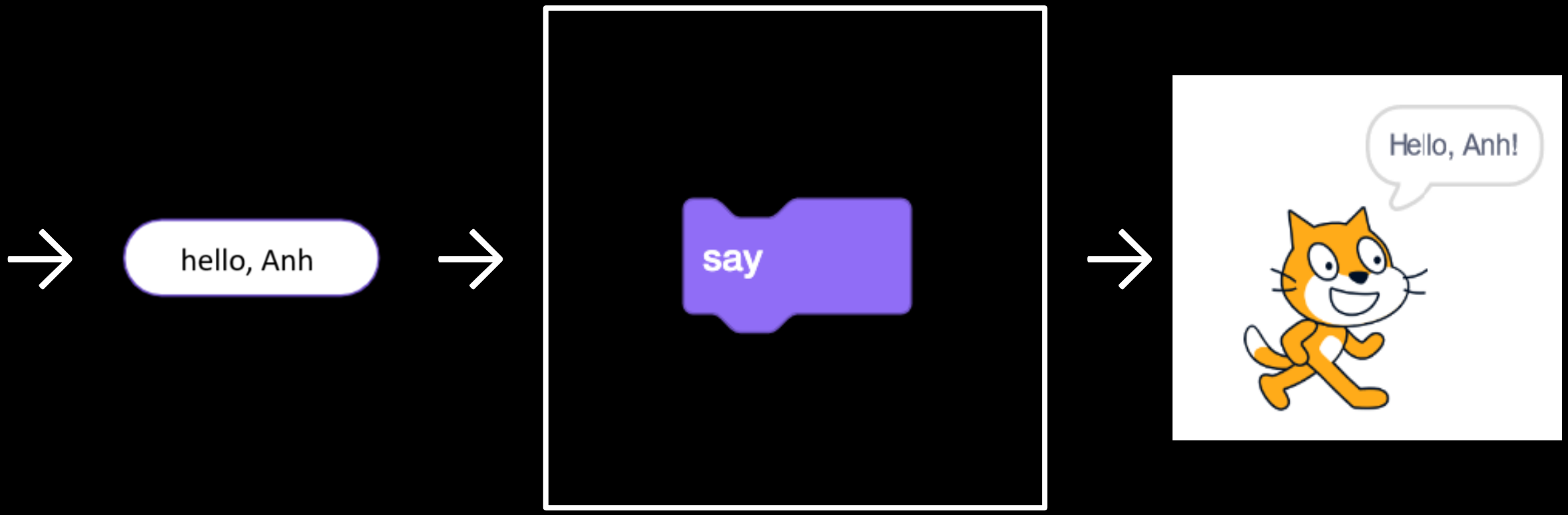

hello,
answer
join
output


hello,
answer
join
hello, Anh


hello, Anh


hello, Anh

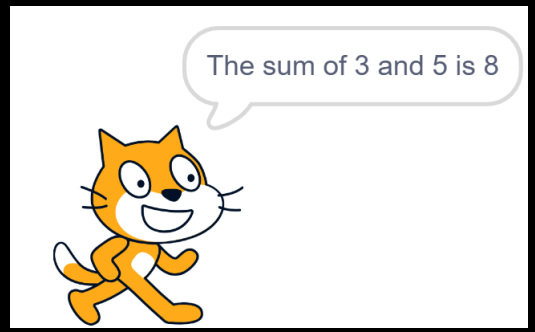

hello, Anh
say


hello, Anh
say
Hello, Anh!

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Exercise 2:

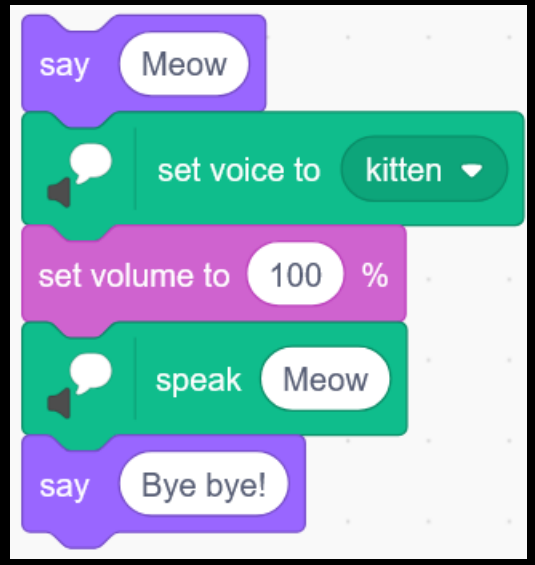
Write a program to enter two values and calculate the sum of them.
Present the result in the format:
"The sum of x and y is z"



Code Structures:

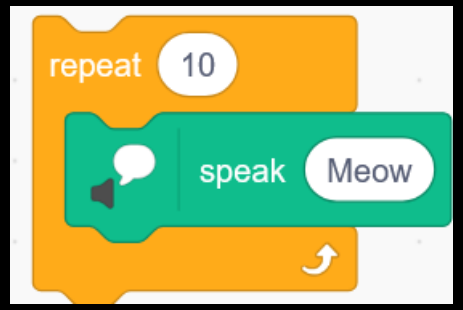
Sequential
Iterative
Conditional

Sequential Structure



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Iterative Structure - Loop



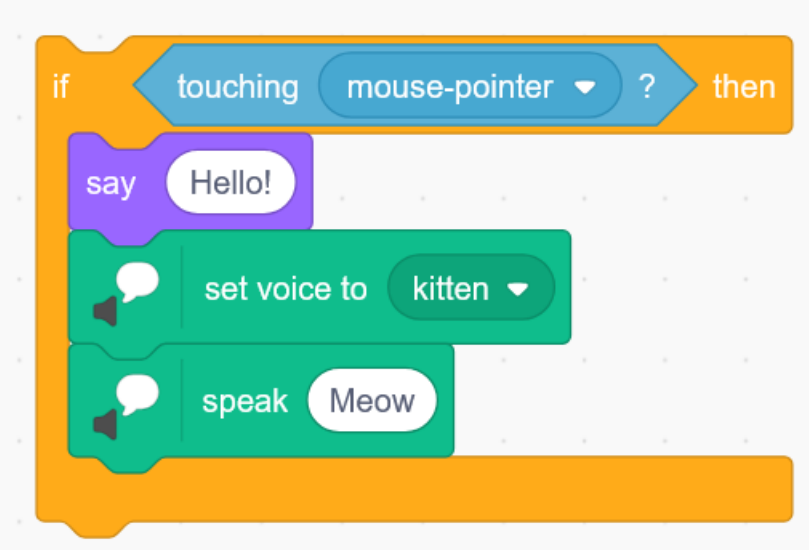
Speak 10 times

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Exercise 3:
Get a number N from the user, let the cat "Meow" N times.

Scratch

Conditional Structure



If touching the cat
then ...

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Exercise 4:
Check the given number is even or odd.

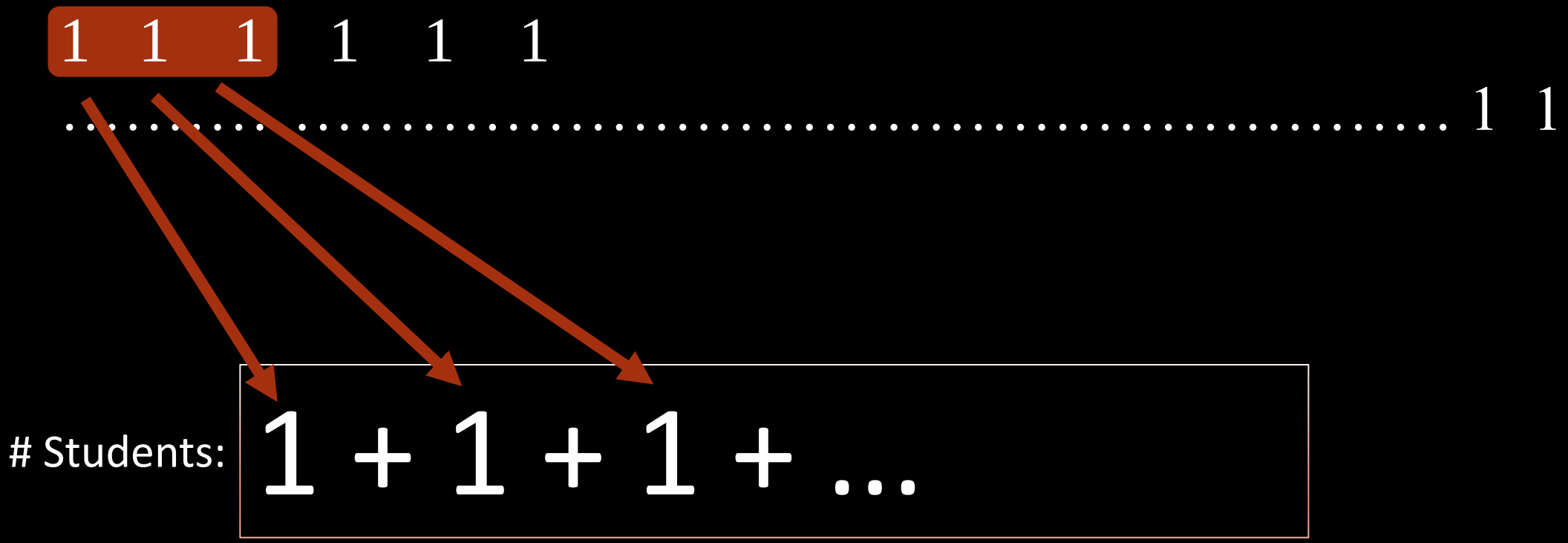
Scratch & Python

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Algorithm Complexity

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Real-world Problem:
Count the number of students in this hall?


1 1 1 1 1 1
. . . . . 1 1
Students:
1 + 1 + 1 + ...

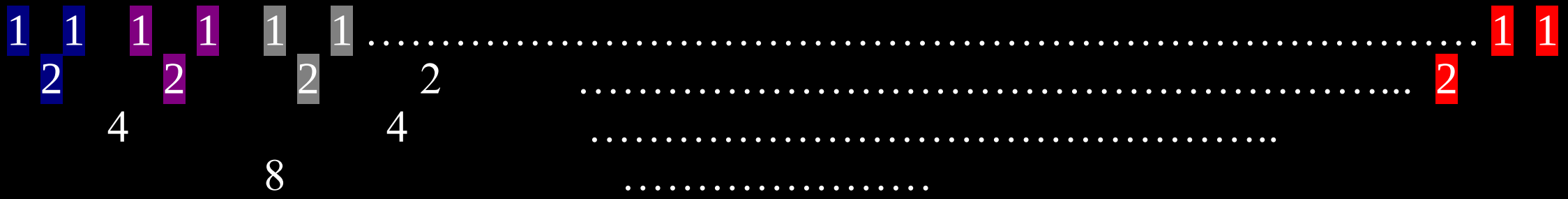
Any Other Ideas?

1. Everyone stand up and think of the number 1.

1. Stand up and think of the number 1.
2. Pair off with someone standing, add their number to yours, and remember the sum.

1. Stand up and think of the number 1.
2. Pair off with someone standing, add their number to yours, and remember the sum.
3. One of you should then sit down.

1. Stand up and think of the number 1.
2. Pair off with someone standing, add their number to yours, and remember the sum.
3. One of you should then sit down.
4. If still standing, go back to step 2.



#

Speed up the counting process!

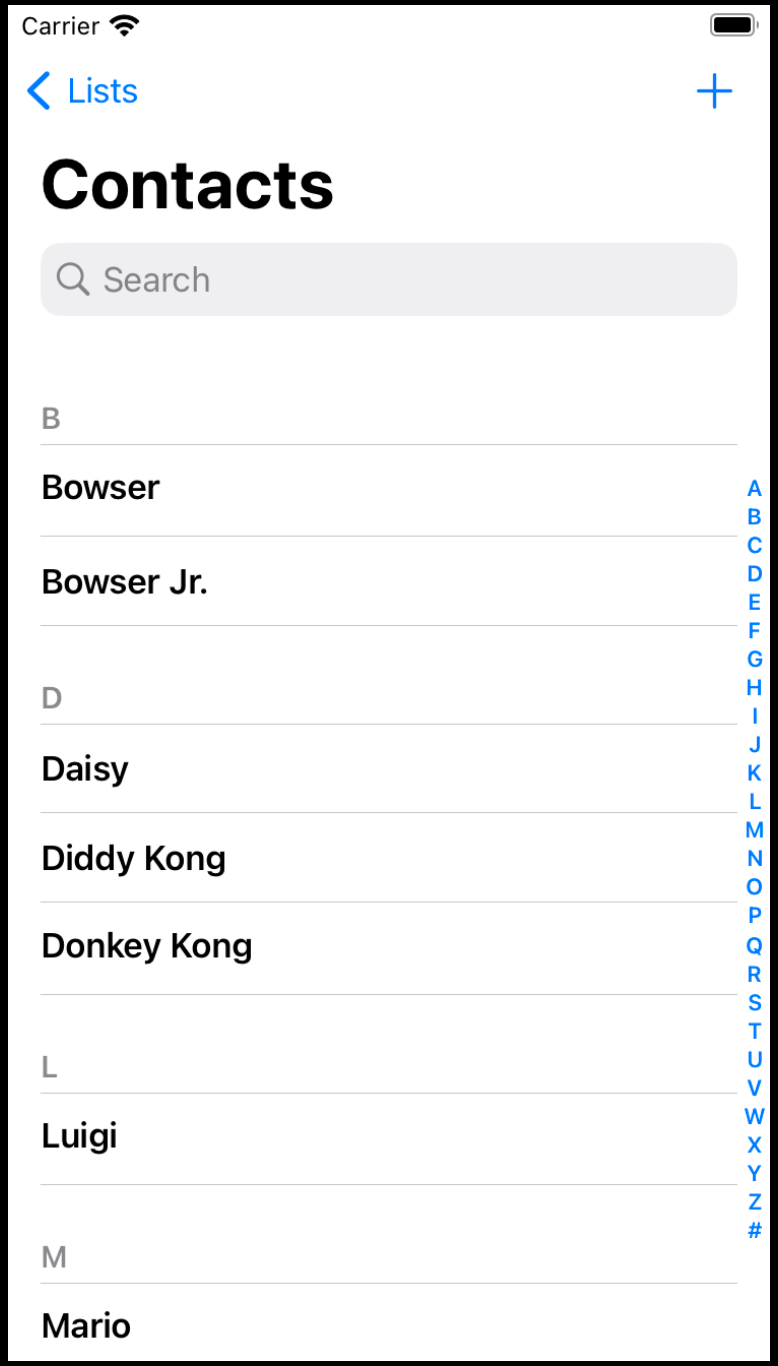
Phone Book Problem - Searching



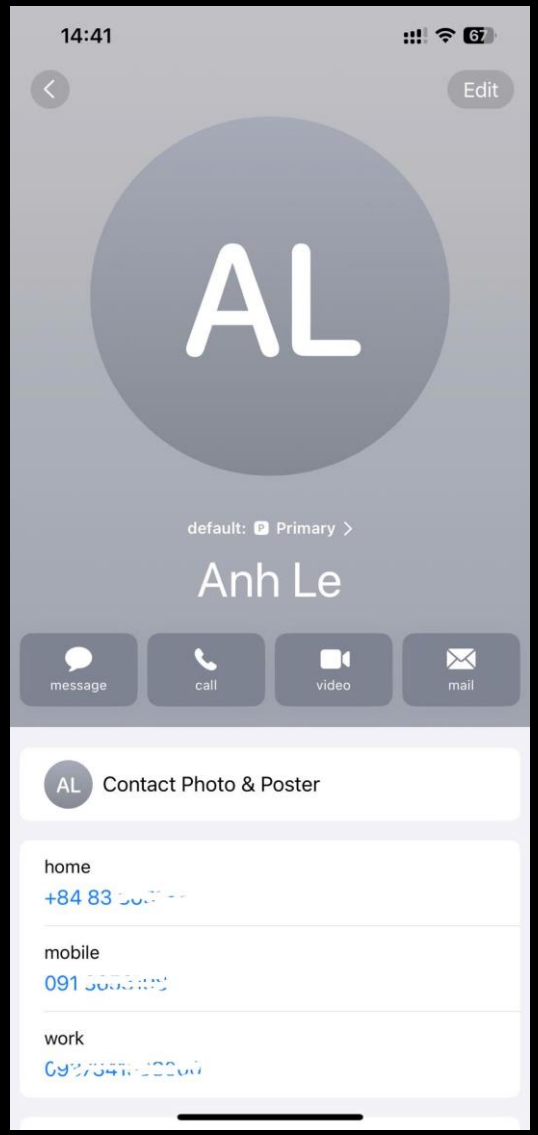


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Carrier
Lists
Contacts
Search
B
Bowser
Bowser Jr.
D
Daisy
Diddy Kong
Donkey Kong
L
Luigi
M
Mario
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z #

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Linear Search Algorithm

```
1 Pick up phone book
2 For each page from the first to the last
3    Look at page
4    If person is on page
5         Call person
6         Return true
7 Return false
```

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Binary Search Algorithm

```
1   Pick up phone book
2   Open to middle of phone book
3   Look at page
4   If person is on page
5       Call person
6   Else if          is earlier in book
7       Open to middle of left half of book
8       Go back to line 3
9   Else if          is later in book
10      Open to middle of right half of book
11      Go back to line 3
12  Else
13      Quit
```

```
1    Pick up phone book
2    Open to middle of phone book
3    Look at page
4    If          is on page
5        Call
6    Else if          is earlier in book
7        Open to middle of left half of book
8        Go back to line 3
9    Else if          is later in book
10       Open to middle of right half of book
11       Go back to line 3
12   Else
13       Quit
```

```
1   Pick up phone book
2   Open to middle of phone book
3   Look at page
4   If          is
5       Call
6   Else if           is earlier in book
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10      Open to middle of right half of book
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12  Else
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```

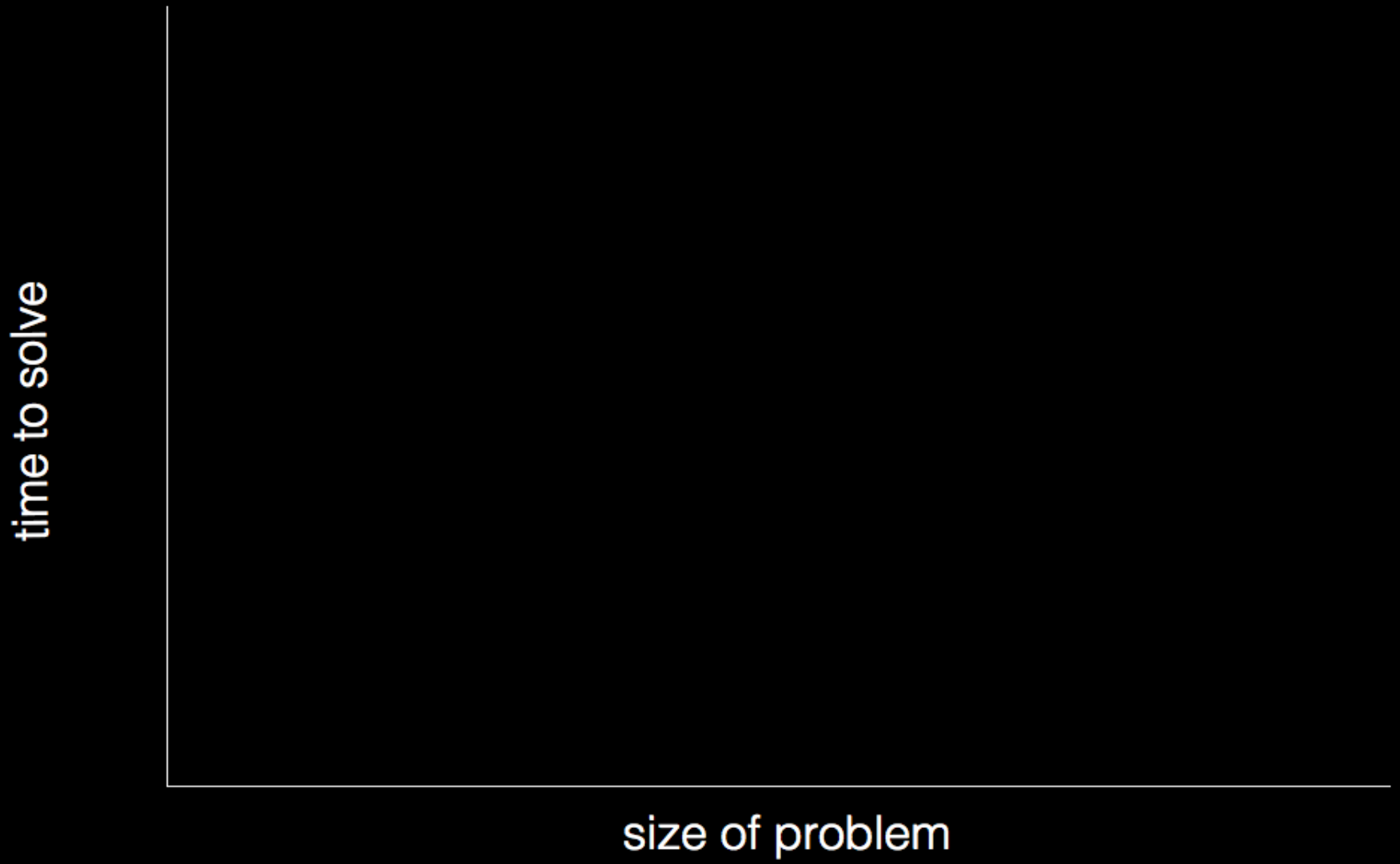
```
1   Pick up phone book
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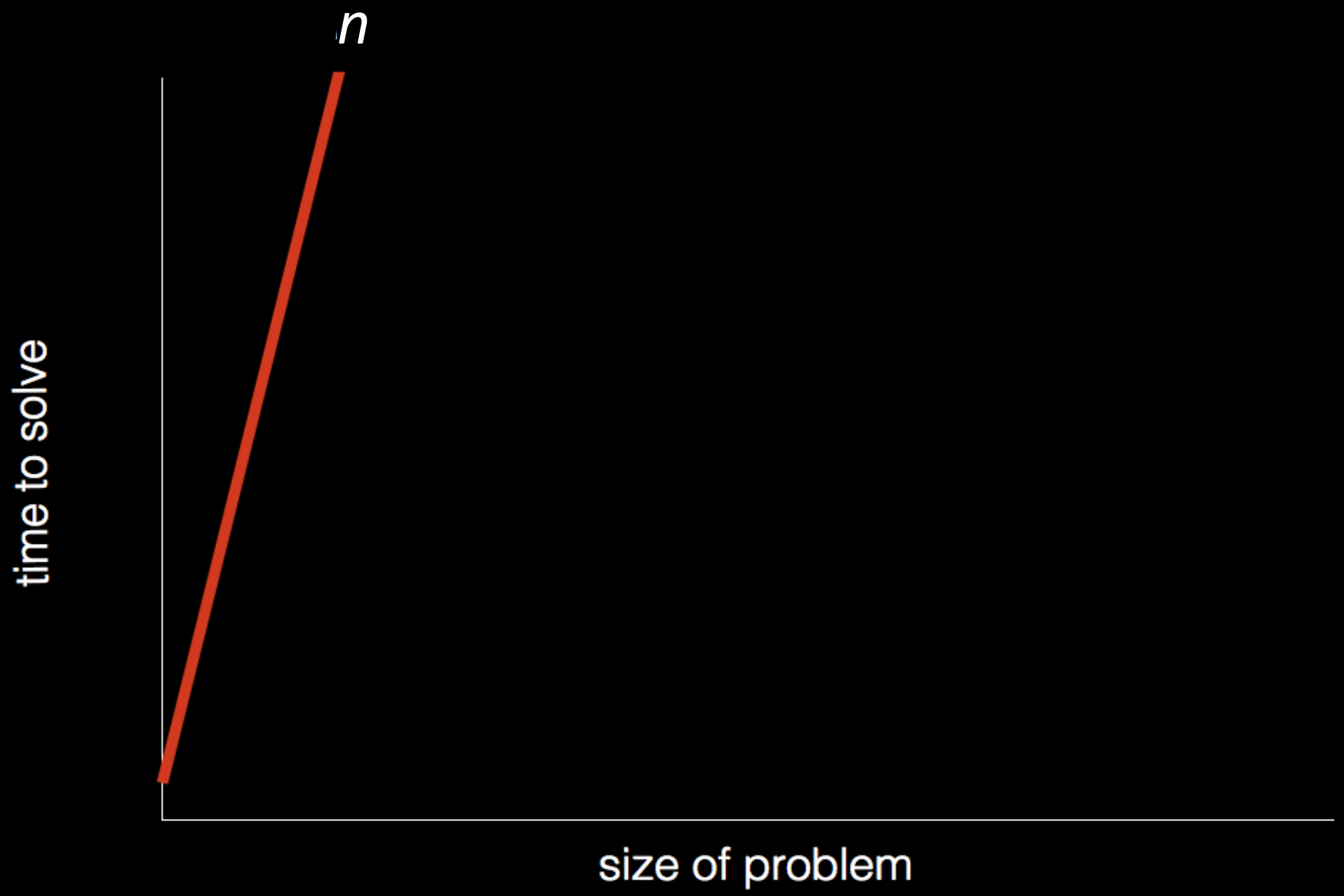
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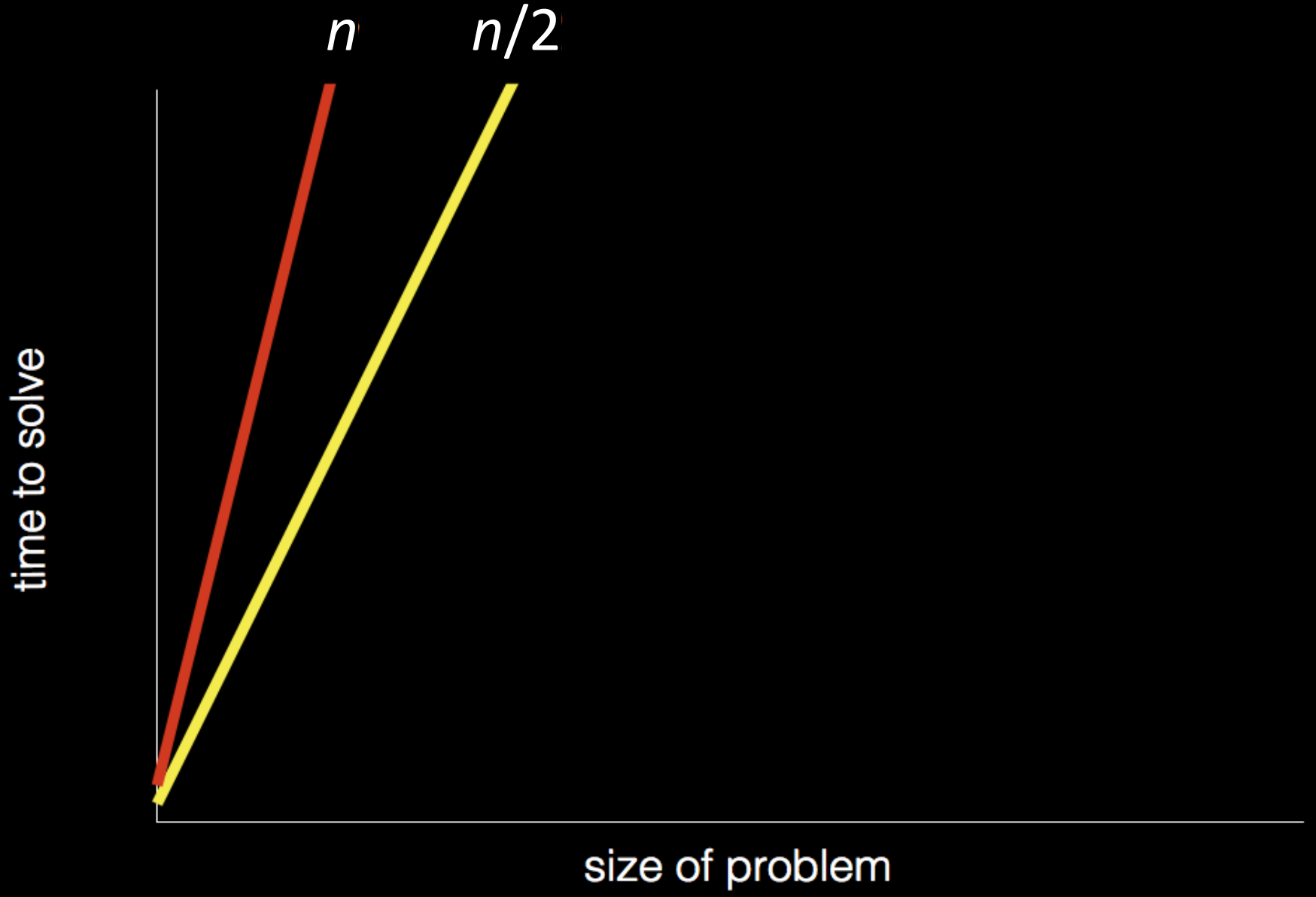
Discussion:
Which algorithm do you prefer?

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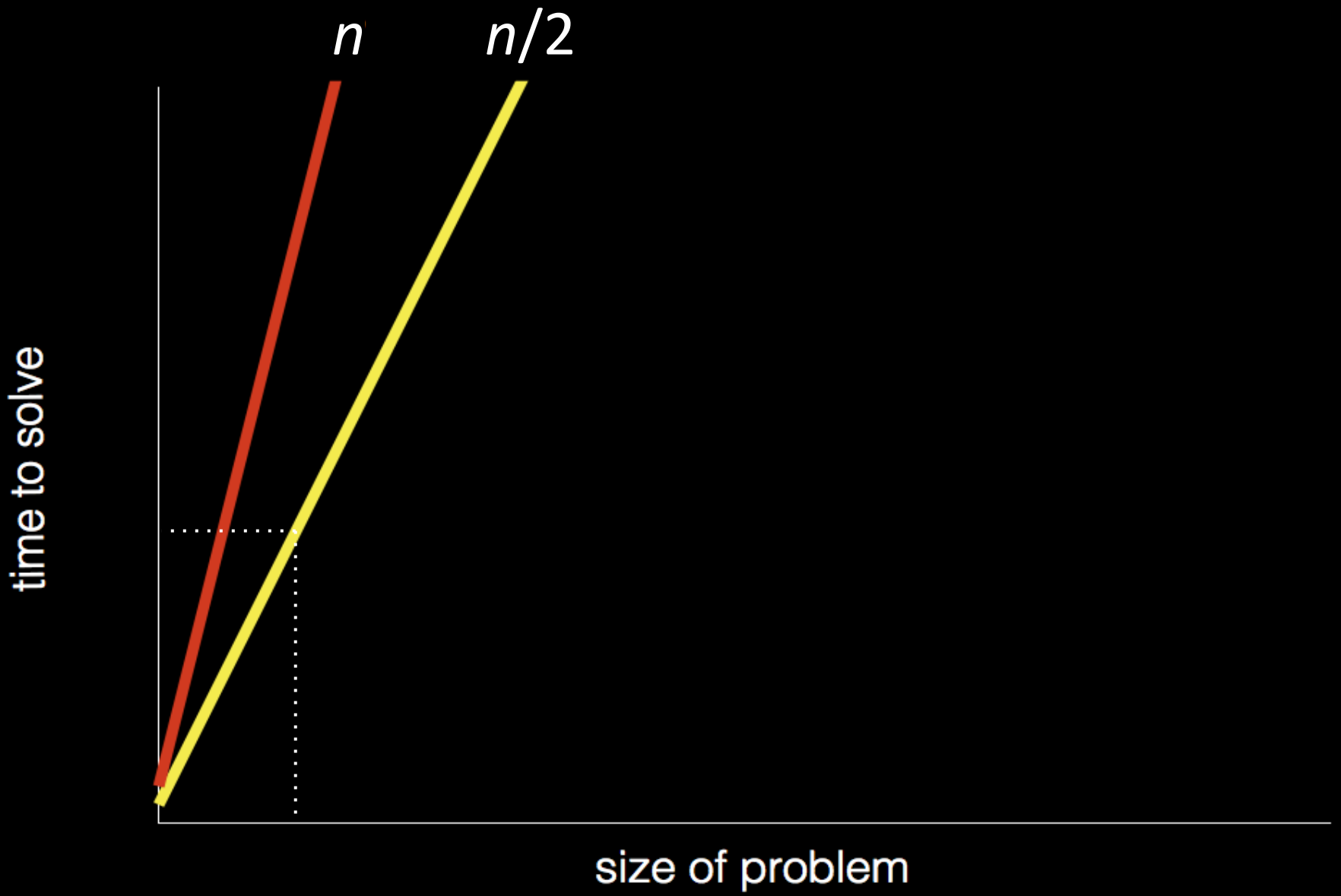
Algorithmic Complexity

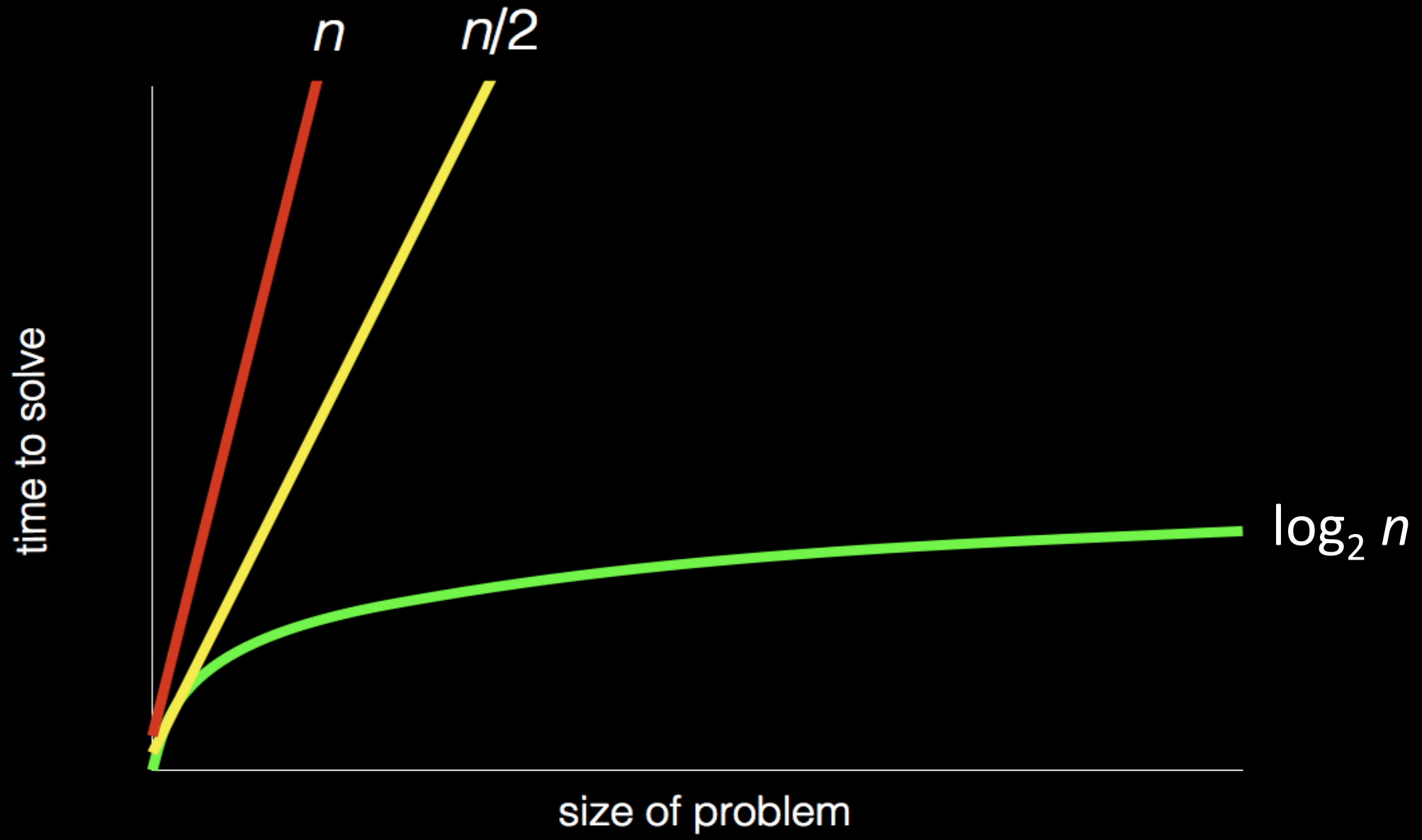

time to solve
size of problem

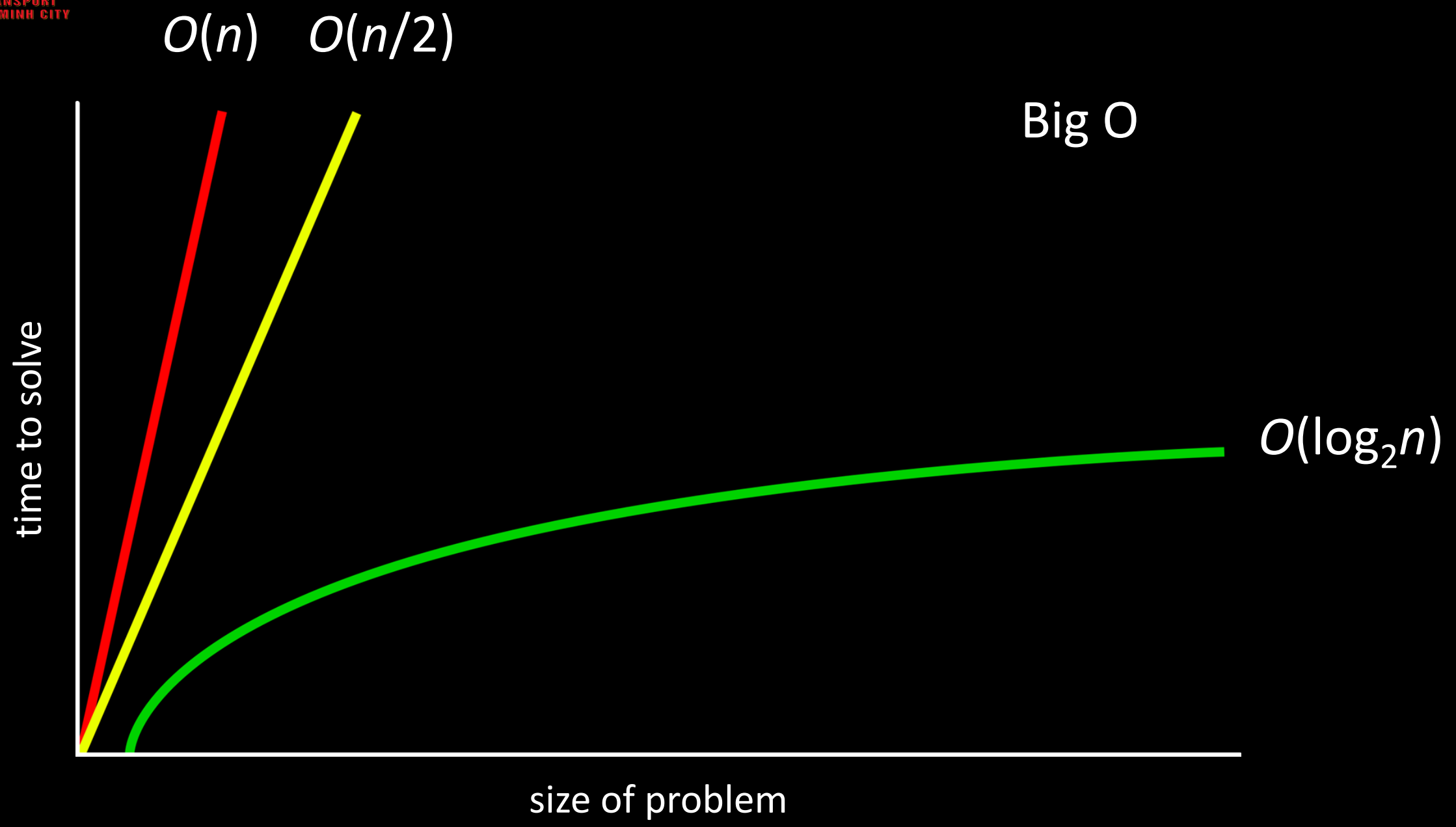

n
time to solve
size of problem

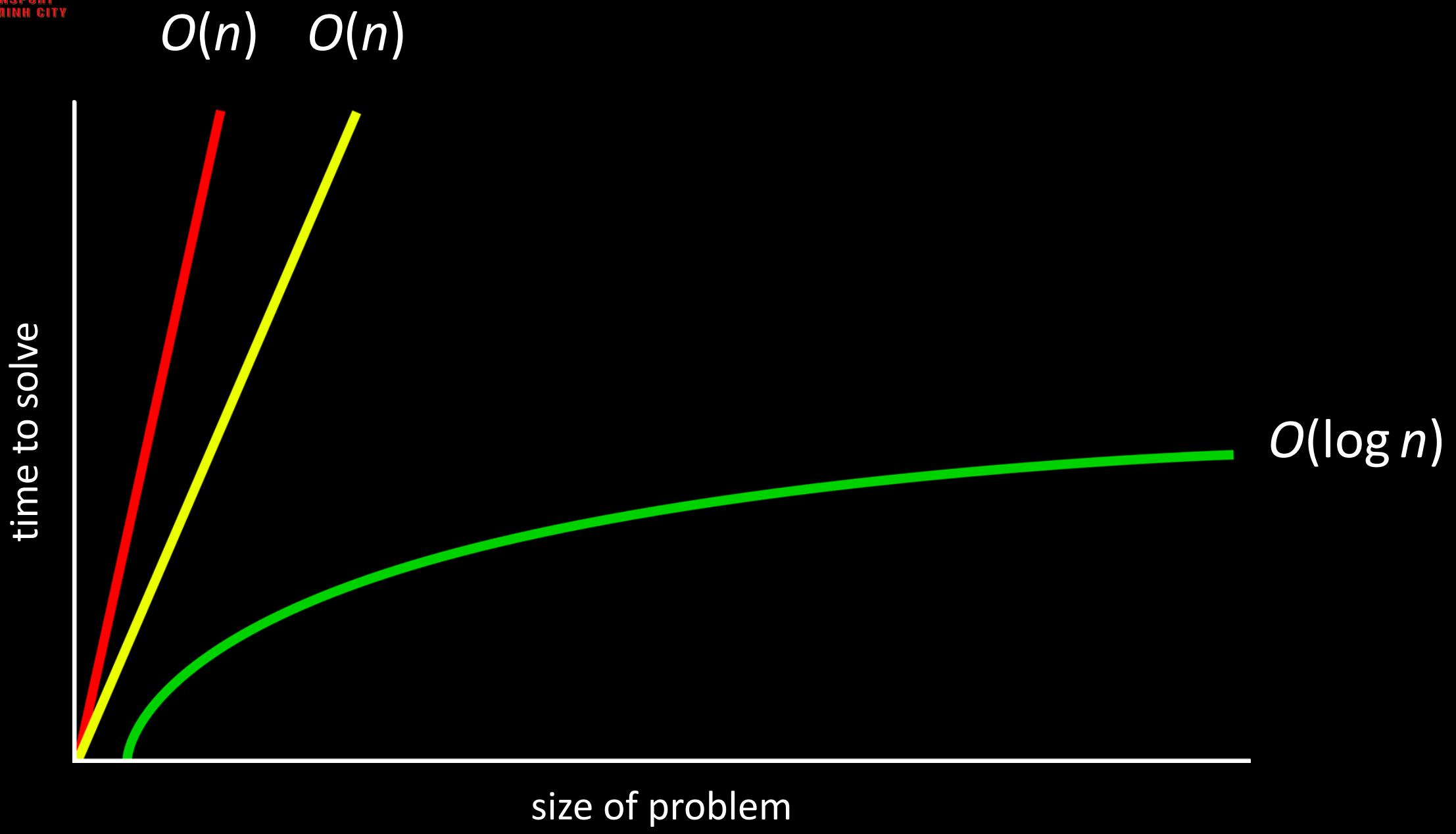

n
n/2
time to solve
size of problem


n
n/2
time to solve
size of problem

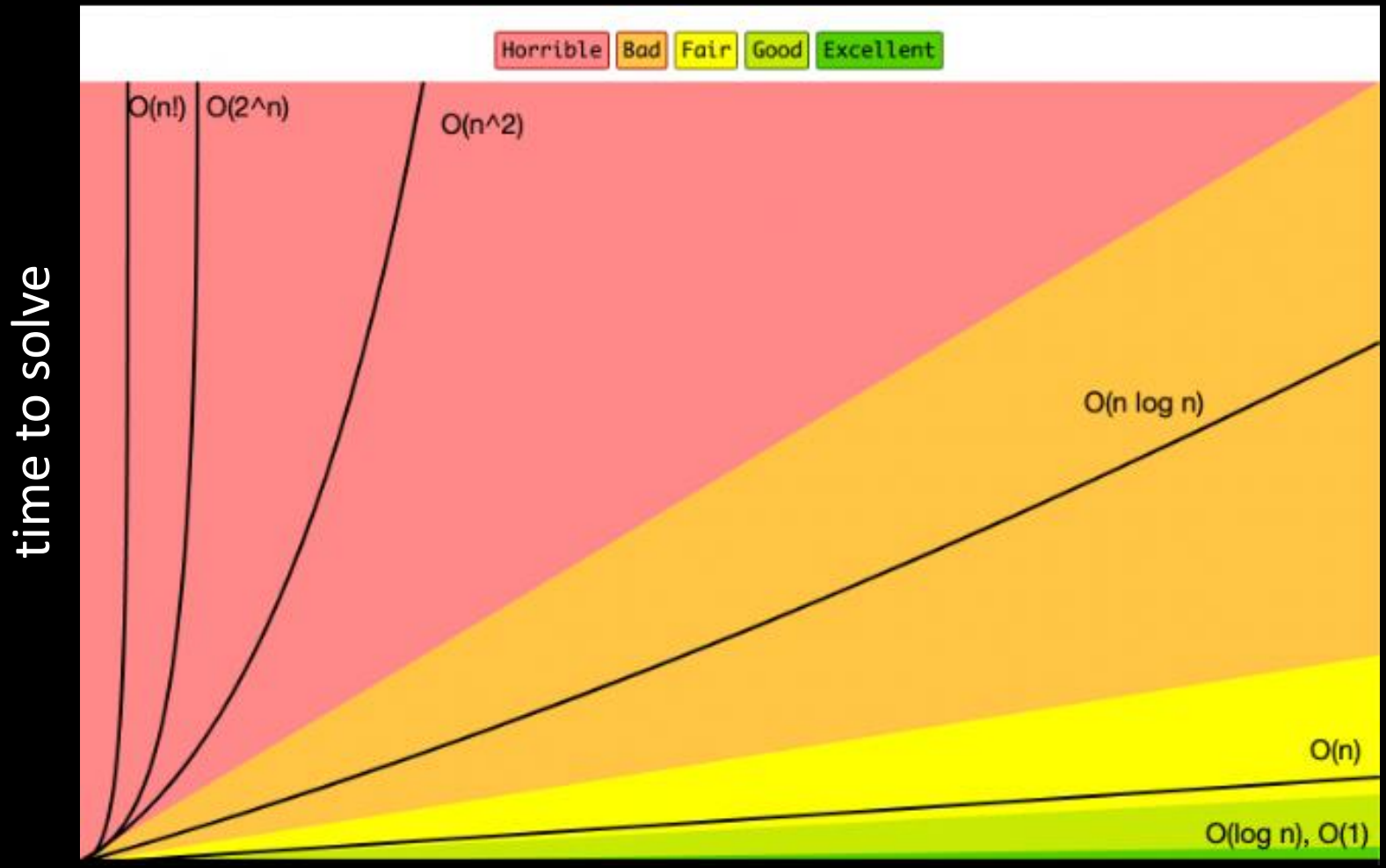

n
n/2
time to solve
size of problem


n
n/2
time to solve
log2 n
size of problem


O(n)
O(n/2)
Big O
O(log2n)
time to solve
size of problem


O(n)
O(n)
time to solve
O(log n)
size of problem

- Constant: O(1)
- Logarith time: O(log n)
- Linear time: O(n)
- Quadratic time: O(n2)
- Exponential time: O(2n)
- Factorial time: O(n!)


Horrible
Bad
Fair
Good
Excellent
O(n!)
O(2^n)
O(n^2)
O(n log n)
O(n)
O(log n), O(1)
time to solve
size of problem

Recap

- Programming – Tell computers what to do
- Control structures
- Think algorithmically
- Algorithmic Complexity